

ActInf ModelStream #001.3: "A Step-by-Step Tutorial on Active Inference"

Source: [YouTube](#) Playlist: ModelStreams Duration: 2:40:56 | Uploaded:

Unknown Transcript method: auto_caption

and we are live hello everyone and welcome to active inference lab this is the third part in a four part series it is model stream 3.0 and today we are here with ryan smith and christopher white and max murphy and we're going to be having a really interesting discussion so thanks everyone for joining just to quickly introduce ourselves before we go to points of process and then into the material i'm daniel friedman i'm a postdoc in california i'm ryan smith um i'm an investigator at the laureate institute for brain research in tulsa oklahoma hi i'm christopher white i'm a phd student at the mrc cognition brain sciences unit in cambridge england hi i'm max murphy i'm a postdoctoral fellow at carnegie mellon awesome thanks everyone for joining for our panelists and the authors for max for contributing your experience and also for everybody who's going to be live and in replay asking us questions this is the third in a four-part series that is going to be highlighting different perspectives and addressing questions related to the active inference tutorial paper of smith at all from uh i guess you could say 2020 but it's been versioned several times and that tutorial is called a step-by-step tutorial on active inference and its applications to empirical data so if you have any questions during the live stream you're more than welcome to type it in the live chat and we'll try to get to it if you have questions after this live stream just leave a comment and we're gonna try to address it as well to learn more and to find out about participating check out activeinference.org that's all so for today in our third session we're going to be picking up with just a brief overview even shorter than the second parts overview and then we're going to be diving into two main sections today ryan is going to be leading a section on the neural process theory and then christopher is going to be leading a section on hierarchical models so two very interesting topics let's just start with one warm-up introductory question before we go to the neural process theory and my question would be let's say somebody were sent just this third part of this four part what would you say is the summary of the first two parts that we can catch them up with right now before we jump into these next two parts so what happened in the first two parts that led us to want to talk

about the neural process theory and the hierarchical modeling today um okay well i can try to start to take a stab chris feel free to add anything but it's obviously a lot to summarize in uh just a couple uh seconds here but in session one primarily we went over the kind of like broad um broad stroke sort of description of active inference um the ideas of how you can do approximate bayesian inference to solve problems with perception and learning and decision making using uh free energy minimization and expected free energy energy minimization um where exact bayesian inference just becomes intractable for most kind of real world problems when you're trying to do something cognitive or perceptual and then we went over how to build specific task models um and specifically we used an explore exploit a simple sort of explore exploit model um similar to any you know a kind of behavioral task that you might use in an empirical study and we just showed how at the level of matrices and vectors and things like that how you specify a generative model in order to simulate behavior on that kind of task and that's more or less what we've covered so far so the kind of general mathematics and motivation of active inference and how to apply it to um to model tasks but that was all like i said at the kind of computational and algorithmic level of description um whereas one of the really nice things about active inference is that you have not only the computational algorithmic level but you also have an attached what's called a neural process theory which more or less corresponds to a specific hypothesis about ways that neurons and neural populations can be connected up in plausible ways to implement the active inference algorithms and specifically the variational updating algorithms that are used to solve the sorts of tasks that you use to model with active inference so that's kind of where we're picking up is now that we've got the now that we can build models and understand their background and motivation you know how can we actually move this into neuroscience and make um make uh trial by trial predictions about what neural responses you ought to see which you could for example feed in um to like an fmri experiment to see if you can actually find correlates of those uh predicted trial by trial responses and just kind of add to that i think the general way that you would feed it into an fmri experiment would just be the standard way that people usually do this stuff so you just put it in and add it as a regressor in um into a glam yeah and as you'll as you'll see we can also do it's not just fmri you can also do it i mean we the the most direct predictions it makes are about earpiece and eeg so and we're going to talk about what the erp is and the biology of it a little bit along the way and other frameworks for how people have interpreted it but yes as we've kind of looked at a few different times there's the observables which are the behavior or the neural imaging whether it's fmri or eeg or some

other type of measurement technique and then there's the measurement of behavior and we're talking about models that start from those observables and then do some interesting generative modeling under the hood and that's where the power arises from so any other comments on that max before we jump into part one with the neural process theory no i'm just uh like i said last time i'm excited to hear about um tying and relating specifically you know gradients on you know variational free energy to you know dopamine expression and just hearing about like the links between those and why that might be an elegant way um to um kind of express things in terms of and what we know about the connectivity of these regions and things like that sounds good then ryan take it away okay so let me just share my screen here um okay so just um as a kind of very brief um you know sort of just catch people up a bit um um so we we started out you know like describing active inference um like i said at the level of generative models um and those uh sorry i'm trying to find the right figure here yeah um and those have this particular sort of graphical structure and just focusing on the one on the bottom right here which is the full you have a particular set of observations over time this $O_{1:T}$ thing and then you have particular hidden states so beliefs about what's going on out in the world outside of the brain and so you have your beliefs about what happened at time one and then time two and then time three and this a matrix thing um is what encodes the likelihood mapping or the belief about what outcomes will be present if particular states out in the world are the case um and then you have beliefs about how in this b matrix thing here about how if you're in particular states at time one how that's going to transition into different states at time two um and so forth and uh this d thing is just an initial prior um about what states are going to be at time one um and then you have policies π_i up here which control which allow the agent to control some of the transitions between states um so for instance the transition of moving my leg up versus down right would be something that the agent could control and then policies are selected based on this g thing which is the expected free energy which is a function of c which is your preferences so the outcomes that you prefer over others so policies are selected that are gonna transition between states that generate outcomes or observations that are as close to preferences as possible um and then policies are also um in part controlled by e which is a prior over policies that acts as a kind of influence of habits and this gamma thing which will be important in the neural process theory i'll talk about which is um it's essentially it's like a temperature parameter more or less on like a dynamically updated temperature parameter that basically controls um how much the expected free energy contributes to policy selection so it's basically saying if i don't trust my model very much then policy selection is going to be

influenced by expected free energy less and by habits uh more um and the way the gamma is updated is through this hyperparameter beta which again i'll uh i'll uh you know we covered this a little more detail last time but i'm just kind of reminding the reader about the elements that are going to correspond to different aspects of the of the um so it just might only happen my only comment on this is again think about incremental layers of complexity in the model on the top left is static perception hidden states observable states and you're doing inference on the hidden states s given your prior d so instantaneous hidden state inference given observables bottom left you're talking about hidden states changing through time with the inference of a b matrix top right you introduce the idea that π policy is going to intervene into how states change through time and that that policy is selected by a preference c that gets transmuted into a free energy g and then the bottom right is the full fleshed-out skeleton where we also see e gamma and beta as described by right so just start top left bottom left top right bottom right and that's adding up one more thing for each layer of the model and then now we're going to be working with the bottom right model yep and um and the the last sort of important thing is is about the the underlying way so each of these equations um on the at each of these boxes um the the way that you infer beliefs about states based on observations is through a set of equations that correspond to approximate bayesian inference through minimizing free energy and expected for energy so the idea is more or less how the system how in the neural process theory does the system settle on a set of posterior beliefs over states through free energy free energy minimization and how in the brain um that's going to correspond to to minimizing particular types of prediction errors um now but i want to be clear though the these sorts of prediction errors are not identical right this is not the same process theory as what's associated with predictive coding and perception where predictive coding is a continuous state-space model as opposed to the discrete state-space models we're talking about that have categories right you can only be in state one state two state three as opposed to these sorts of continuous state spaces associated with things like brightness for example um so so this is this has a similar kind of prediction or minimization flavor to predictive coding models but it's not the same thing um so now um to start with um kind of abstractly the the way to see how prediction errors come in really naturally is just to remember that free energy right one one decomposition that we've talked about in previous sessions is you can think of free energy as to be being decomposed into a complexity term minus an accuracy term um where accuracy just means basically how close your predicted outcomes are to um the observed outcomes and complexity corresponds to basically how much you have to move your beliefs um from prior

beliefs to posterior beliefs so minimizing free energy means coming to the beliefs that um change your beliefs as little as possible while also maximizing accuracy um so so that's the kind of idea and a very kind of intuitive way to think of free energy and so given that definition you can think of the gradients or the way that free energy changes with respect to these approximate posterior beliefs that we're trying to come to right these can always be expressed as a particular mixture of prediction errors and you think about this is pretty intuitive because complexity like i said is just the average difference between posterior and prior beliefs so minimizing that difference can be seen as a type of prediction error and minimizing and then accuracy is similarly just the difference between predicted and observed outcomes so minimizing that difference can also be seen as a as a type of prediction error so active inference can be implemented then um as this kind of prediction error minimization that just corresponds to a minimization of these two differences at a computational level of description as opposed to an algorithmic or neural level description so the the kind of standard and fairly daunting looking figures that you'll typically in active inference papers associated with the neural process theory looks something like this so on the left you have a bunch of initially pretty scary um you know equations that correspond to the the actual kind of dynamics that a neural system is going to have to implement um and then on the right these kind of little fun kind of schematic ball neuron setups where you have particular layers that are supposed to kind of um you know kind of heuristically perhaps correspond to different um different layers of neurons and cortical columns um and each of these layers of neurons are sort of proposed again very just kind of like schematically to correspond to different variables in the equations over here and the synaptic connections correspond to particular mathematical operations so um so here kind of the easy way to think about it is that these uh red connections here are excitatory or excitatory just means addition in the equations whereas these inhibitory uh blueish purple connections with the balls at the end instead of the arrows are inhibitory which just means subtraction in the equations whereas the green modulatory connections um that just corresponds to multiplication um in the update equations on the in the other side of the panel so the idea is that you can use this kind of using this kind of scheme you can just follow along you know just follow the synapses and the ball neurons that correspond to the particular variables um and you can kind of get a sense of you can just connect up neurons like this to solve the equations that need to be solved to implement active inference just say one thing about this i think makes it a little bit more intuitive so because all of the updates are done in log space addition and um face click adding or putting probability distributions together is

just addition and subtraction which is nice thinking about in terms of this scheme and then generally speaking the modulatory connections will correspond to some precision parameter which will modulate the precision of a distribution if that makes sense so that's kind of how you can think about the rule or i find that helpful in thinking about the relationship between something like this and the actual update equations themselves yeah no absolutely uh yeah thanks for adding that it's almost like the anatomy of the model we're doing a schematic neuroanatomy that's simplified or abstracted based upon the actual patterns of neural connections and then we're looking at the anatomy of this computational model and thinking about how it could be plausibly carried out on neural uh systems neural processes but also it stands alone and so this is just another way that we can look at it yeah so so another thing that i really want to be clear about is that active inference per se isn't committed to this particular implementation there's there's quite a few different ways that you could come up with to connect a bunch of neurons together to solve these equations so there's there's not kind of a one-to-one mapping between the way that the equations look and the way that neurons could solve those so this is an example but ultimately it's a separate empirical question what way uh the brain is actually set up to solve these equations even if those equations are what are being solved um yeah exactly and even if so then we can also let's say it turns out variational message passing or like marginal message passing is just a really bad metaphor well that's fine because we've got all sorts of other things like belief propagation blah blah blah and they can all i think the main thing is that you're minimizing variational free energy and expected free energy the rest is part of specific parts of a process theory that we can hopefully falsify or at least more realistically not find useful anymore yeah so let's just sorry one more note on that you brought up how there's the computational and the algorithmic levels and we're in the process theory realm so for those who might not be familiar with process theory or mars levels of analysis the computational level is like the function like you have to sort a list from the biggest to the smallest you can't falsify that idea it's computational it's functional then you go one layer down into the algorithm so you say i have a sorting algorithm there's many sorting algorithms but this is the implementation as it's actually going to be written in a certain pseudo code so that's one of many possible algorithms that do a certain computational task like sorting and then the process theory is like the specific program you can think of it's the actual implementation now as to whether it's actually being implemented that way you might be right or wrong so that's what we're talking about with the false falsification of the process theory but at the computational level it's what christopher just said it's the imperatives that there's a free

energy minimization happening and now we want to be building up specific plausible examples to work within that framework for just to jump in real quick it sounds like in the next stream we might be discussing more of these empirical examples and instantiations but just to say that from the bio biological standpoint of what's going on for example when we talk about a modulatory layer or that that could happen in a multitude of different ways um in in terms of you know changing what the membrane potential of the cell is and making it more responsive or receptive to being activated or it could happen through you know gradual um plastic processes that recruit more receptors to the um to the surface and cause it to become more responsive to any individual message so it's really going to be context specific and depend on the experiment and what um circuit we are specifically studying uh yeah absolutely and i mean the the thing you mentioned about plasticity uh makes an important point that i yeah i should make sure that i you know i might have forgot to mention but but so in the context of inference right over the shortest time scales we're just getting observations right and based on the way that these the current weights of all these connections the system is just going to settle on a free energy minimum right like a predictionary minimum and then the the activation patterns in these neurons specifically these ones at the top here the $s_{\pi t}$ and s_{σ} or s_{pythau} and s_{tau} are gonna be the posterior beliefs about what states you're in um and uh but that's again just given a fixed set of synaptic connections that corresponds to your generative model now um but with learning like we discussed last time when we shift learning simulations learning corresponds to slowly changing the synaptic the strength of the uh each of these synaptic connections themselves um so you also over slower time scales minimize expected free energy and um through um through changing these connection strengths um again over slower time scales via kind of synaptic plasticity processes um like long-term potentiation and long-term depression for example um as a way of improving the accuracy of your generative model um so so just to kind of reiterate inference corresponds to changing activity levels whereas learning corresponds to changing synaptic strengths um so if that's if that's uh helpful um so so again yeah i mean like like they were saying you have basically three different levels of hypotheses to test one is at the computational level does the brain uh minimize free energy and expected free energy or at least is it helpful to describe it as doing um and then second question is given that that first one is true which of several different algorithms is the brain using and then question three is given that it's using a given a specific algorithm how are the how is the brain set up to uh implement that algorithm so there's kind of a one-to-many mapping between each level um and a final point to make is actually some

people talk about these different you know are called quote unquote marion levels of computational level algorithmic level and implementation level people talk about them as though they're entirely independent um but they're actually not necessarily independent in all cases because sometimes the algorithm can depend a bit that can be dependencies between the algorithm and the implementation because some of the things that the algorithm is going to have to compute are things about for instance like energy costs which are going to depend on the implementation um so anyway just as kind of a side point but um but in general um just to kind of you know walk you through you know a sense here so you might start out with these observations at the bottom right and these are going to go up through these excitatory connections to this pink layer 3 here which is going to correspond to the state prediction error we'll talk about so these excitatory connections are going to drive increases in activity in these pink guys and those are going to be inhibited by prior beliefs about states at this higher level which jointly right is going to lead this uh this level to decrease while the system finds states up here at the level above that best account for um or predict those observations um so these red arrows going up here um can correspond to interactions between a and o in these equations um and so you can kind of you can walk your way through kind of step by step and we do this kind of step by step in the paper where each of these interactions corresponds to elements in the equations so for instance like to get states to get states up here right that corresponds to this fourth equation down and that's by multiplying policies um by states given policies right so π here multiplying the the s's and the t's here that are added so we'll lead to the posteriors over states just as one example um and then another thing that's important to um to stress is a lot of times you'll you'll notice in these uh diagrams that there are kind of multiple uh rows kind of going back right so there's like these three pink ones and then these three other pink ones behind them um and that's because um each of these uh posterior beliefs over states the system is trying to come to um are with respect to a particular so the thing is actually calculating what its beliefs would be under each of many policies where each of these rows corresponds to a policy and that's what allows it to then infer a distribution over policies based on the states that are expected under each policy and the outcomes that they would be expected to generate so so again i'm not going to kind of do this for you now just because it would take forever but if you just kind of look at each of these equations you know where addition is where multiplication is etcetera then you'll find um that these connections um you know basically reiterate or or show uh uh how this can implement uh how the neural connections can implement each of these equation um but um to uh to kind of go into certain aspects of how

this would work and where the neural processor really really comes in in terms of making um empirical predictions just just just to jump in real quick um for the sake of moving forward um just to help me thinking about what the biology is as we start to talk about the neural process theory here you have um pie and you have kind of like you know connections that are inhibitory excitatory um sometimes when i think about phenomena in in the biology you know you have an emergent concept that that's the result of kind of a population of cells in this case would it be um better to think of this uh like a policy as the actual like you know the activation i'm looking at a single neuron or am i going to be thinking about that like at a you know a layer or a population level oh yeah yeah very very good point i mean like i said these are very schematic and in any active inference paper you'll see that you know the statement as the cheese balls is supposed to correspond to a population of neurons okay right so a population where i mean right like one subset of neurons in a given population might encode the expected free energy of each policy right so activation across the population of neurons would code for instance a distribution over um all the policies being considered um or something something along those lines so yeah no no no one's no one's thinking about this as uh each of these being you know single neurons that are solving um all this i mean that would be um uh really it'd be a really vulnerable system very little damage that system would uh would knock the whole thing out um so um so so yes definitely they're they're supposed to correspond to populations but a lot of the details of that is really um kind of underspecified in the in the theory as stated it's still fairly abstract um kind of at the level i'm showing here so uh so is that is that that helpful yeah yeah that does okay um so okay so the first ones here to focus on the left are so this epsilon pi tau thing um and this is what's called state prediction error um and more or less what this does is by minimizing this we end up with a posterior over states um and then the actual variational free energy for each policy is just calculated based on beliefs about states and that error signal um so and this corresponds to layer three um and layer two here um in the um in the uh schematic on the on the right um so to um to give you a sense so in the paper we show a specific example like a worked example of calculating this kind of prediction error um and so what this says so this is the prediction error for each policy for each time point tau and the the first part here this one half ln b pi tau s pi tau minus one et cetera all the way over the whole thing here with the b's that are multiplied by one half you can think of this whole entire thing as your prior expectation so b from pi tau minus one um this is saying what do i expect about my current state given the state that i was just in before um whereas this one is something a little more interesting it's saying um this is saying sort of beliefs about um

expectations that come from the future so you know what are my beliefs about a state at a given time given my beliefs about the state after that um and um you know this this might seem a little bit counterintuitive but um it allows for uh retrospective inference um which is something that humans definitely do so think about think about a case where you're sitting in a room and it's dark and you don't know whether the room is the green room or the red room um and but you sit there for like five minutes and then all of a sudden you turn on the light and you see that the room is red you don't just infer i'm in a red room now you infer i've been in a red room the whole time so your beliefs about the past are updated based on your beliefs about the current time point so what this this second be here so expectations propagating from the future to the past allows you to come to beliefs about the states you were in at earlier times based on beliefs about later times um and um and so so it's important to also reiterate here that each of these s's um s for pi you know s pi tau minus one um these beliefs about each s at each time point are updated at every time point so at time three i can observe something and update my belief about the state at time one um so this is kind of being recomputed at each time point with a new observation but about each time point um you know in both the past and the future um as well as the the present so so broadly then this whole thing this one half times both of these b's and s's is your prior expectation and then that gets added to the likelihood function here which is so a um times o which is the actual observation um at a given time and so this is like your uh this thing together is like your likelihood um so this again is very similar um to uh just kind of another form of representing uh bayes theorem where you have priors and you have a likelihood and you're trying to come to something that will inform your posterior beliefs over states and then that whole thing is subtracted from beliefs about the current state at that moment and more or less if you just kind of run this over and over again you'll settle on a set of beliefs about states that minimizes um this thing um and just to show an example of that um we specify here um you know what a could look like right so in this case um state one versus state two the two columns state one would generate say observation one um with point eight um whereas state two here right and the right column would generate outcome one with point 0.4 and so forth so that would be your likelihood now b here like i said before this is your prior from based on the previous time point and here we're just saying column again here is state at time tau and rho is at time tau minus one or uh you know from tau minus one or this is basically this is saying given that i was in a given that i was in a state at time tau what is my belief about the state i'll transition to so this is saying i believe if i'm in state one i'm most likely staying at state one but there's a point two probability that i could move to state two um

whereas the μ be π tau is point three point three and point seven point seven which means no matter what i expect it's more likely i 'm gonna move to state two μ and then what i actually observe is outcome one μ and my prior beliefs are just 0.5.5 so i have no i didn't have a strong prior either way to start with μ and this v thing is μ just a depolarization u variable which doesn't come into this u h predictionary equation well so i 'll just kind of leave that for below but so this is just an example of how if you plug in each of these things into this equation then just kind of follow along with the numbers eventually you'll end up with this prediction error this negative point two two three one negative point nine nine nine one six three μ and u h and you can think about so then if we move u h here then we can take this v which is your initial kind of depolarization μ u h level in a given neuron then you add the prediction error to that μ so the depolarization depolarization level changes so this would be a change in the activity level of the neuron in terms of but then posteriors over states then correspond to normalizing putting a softmax over this which then turns it into your actual so this is b so this ultimately says my posterior belief is that i 'm more likely in state one you know 0.66767 then in being in state two point three through three three μ and this then would correspond to the firing rates μ in the process theory μ associated with μ u h each possible state so v is to be your depolarization μ and s is going to be the resulting yeah just add something to this so generally thinking e here or sorry not e epsilon yeah yeah a little epsilon is a free energy gradient μ it is a variational free energy and then this also what would what we end up doing here when we're adding essentially when we have v when we're plussing our free our negative friendship gradient we're essentially just doing a gradient descent on variational free energy that's exactly what this is actually and that's how it's implemented in the code the only difference between this and what we're doing in the code is that we have little steps have a little step thing so we divide our epsilon by like two or whatever our time constant for gradient descent is so when you know tying it back to that previous u h schematic that we had where we had the arrows that were excitatory and here we're adding this on to the putative you know membrane potential that we have described here that might reflect these kind of probabilities then if we have this kind of perhaps this if this were an excitatory interaction that we were looking at then maybe if that excitatory interaction from one particular layer to another was representing that u h variational free energy gradient we might expect to see this kind of response and we might measure that in terms of some kind of related you know u h bio like biomarker μ of interest is that more or less than yeah i mean so well i mean just to go back to the kind of schematic here like where this is happening is like

right here right so you have b from time one right which is this purple guy so s through b from the previous time point adds on to the prediction error at the current time point and s from the future belief about the future also adds here whereas the likelihood a um yeah is added from from oh uh there's this inventory thing here um anyway i mean so that's that's the uh i have to walk through a little bit to figure out where this uh inhibitory what this inhibitory one is coming from um oh this is the minus $\ln$ yeah the current state yeah um yeah okay but anyway so that's that's kind of as an example um but uh but yeah so so by doing that you get this posterior over states that gives you both uh predicted depolarization levels in cells and predicted firing rates um that being said um those and of themselves are not that easy to measure empirically unless you're doing like single cell recording or something like that so the idea that gets taken sort of from here is how you can get from these predicted firing rates to something that you can measure um like event related potentials in um doing eeg or electroencephalography where you have a bunch of electrodes basically attached to somebody's scalp and you're measuring changes in neural activity changes in potentials through the scalp from the brain when a person is exposed to a particular stimulus and the uh the neural process theory uh the kind of prediction is that um the erps are gonna correspond to the rate of change um um in beliefs uh which is gonna be the rate of change in um where's this slide the rate of change in um in v very i've changed this depolarization thing which is going to be the prediction error so so just to show an example of this um that me and chris uh published recently um in the uh literature on the um on visual consciousness um you know there's a big interest in in what erp's um are specific to having conscious experience um and an example of an erp that in the past has been linked to conscious experience but more recently there's been some sort of controversy because it looks like you can get these p3 potentials in a way that's not specific to conscious experience um and so we're interested in seeing whether the neural process theory would allow us to kind of show um where this is coming from and could reconcile these sorts of results and so it was based on this really kind of simple paradigm where you start out the person starts out looking at a stimulus like this where there's just these red discs and these um kind of slanty lines um and then in some trials what will happen is they'll kind of transiently be a little square here that comes up um in the midst of these slanty lines um then it'll go away um and basically what what um what the manipulation is in some cases you have the person you know focus specifically on these red discs on the outside to see if they change color and then you ask at the end you know did you ever notice these square things that pop up um and there are cases when um when they do see the square

in cases when they don't and you're looking for differences in erps corresponding to when they do versus they don't under conditions of different sorts of attention um and so what so what we were trying to do is to see whether um when you built um a an active inference model that um that corresponded to this particular um attention uh task um that's called an intentional blindness task um you know could we reproduce the the p3 pattern so this kind of in red here this extra um dip um in uh neural activity as measured by eeg that um looks like it typically only happens when a person consciously experiences something but with these recent exceptions where it looks like it depends just on whether a stimulus is a task relevant or not um and chris actually i think you know this was primarily your work so so maybe you could describe this in a little more detail yeah yeah so first of all i think that just give a shout out because we've got to reference it on the slides is um this paradigm was designed by michael pitts um and he has done a lot of really incredible empirical work on um visual consciousness so just kind of important to acknowledge that anyway um so generally they speak there's three phases in this experiment so phase one is they're not told there's going to be a square kind of self-assemble and they're asked just to monitor the disks on the outside and then there's a huge literature on this but generally speaking in intentional blindness when you have a setup that's kind of analogous to that about 50 percent of people are intentionally blind to whatever your so i think the really famous one is like a group if you ask people like monitor um how many times a basketball is being passed back and forth between a group of people a gorilla can walk past and wave at people and about 50 of them um don't report seeing it um so the idea here is just kind of operationalize that a little bit more carefully um and then in phase two is after everyone's kind of been because like literally when you ask people after phase one hey did you see a square you're kind of alerting to them the fact that there was a square present there but it's still not top the square's still not task relevant so in phase two they just do exactly the same tasks monitor the external disks um and in phase three the manipulation is they then say okay well we now want you to count how many times like hit a button whenever the square appears and so the idea was was that if you kind of see here when we say percent scene we basically said a hierarchical model which we're going to look at in a minute and we just had it kind of report what type of trial it thought it was in did it see a line or not where a line was like this categorical variable or did it see a square um and essentially we just in the same way that you might in psychophysics like titrate the stimulus or the contrast of a stimulus or whatever until people see it 50 of the time basically what we did was we titrated it until the model saw it 50 of the time or titrated the precision of the a matrix kind of corresponding to contrast or

whatever and then we just manipulated attention where attention corresponds to basically how precise the model's a matrix was um and what we showed was that when attention which we operationalized was basically like we had two steps of attention we had focal attention and um like super precise task relevant attention when there was just the focal attention condition the model reported seeing the thing 99 of the time so basically all the time and when there was precise like task relevant attention the model reported seeing 100 of the time again but we see a dissociation between the presence of the p3 in line with empirical results and so the idea so i can actually just highlight this since i'm the one with the cursor so so in this first condition so this is when uh like chris said the you know the thing had no expectation right that it was gonna that there's gonna be any square there and um and the precision of the a matrix was low because there was no attention um so as a result the erp is totally the simulated erp is flat and similarly this uh the second the second dip here which is what would correspond to a dip here is absent and here at the bottom are the simulated firing rates so in the first level basically when black at the bottom changes to black at the top that just means it starts to believe the upper row is a belief that the square was present and the bottom is the belief that the square was absent so basically what this means is when it goes from black at the bottom to black at the top that the belief changes so it's more likely that the stimulus was uh present um and then but you can see because the a matrix is uh imprecise then um which corresponds to like a low stimulus strength and this um then this is not very uh you know it's still pretty blurry and then at the second level these are beliefs about sequences that are what's actually gonna in other words did it go from no square to square and back to no square um and you can see that the beliefs remain really imprecise um because it's not uh it's not attending to it it's not task relevant um in this case though like chris was talking about you can see that even though it's not task relevant the beliefs at the higher level ultimately do converge onto this top row which is the one where the square was present but you can see that the beliefs change pretty gradually whereas when attention was precise because it was task relevant then you can see that the belief changes from being kind of flat over these two possible top states to being black fully confident that the square uh turned on and off and that really quick change in beliefs is what the strong erp and in this case because of the higher level it's a kind of late erp corresponding to the p3 so really what's going on is the prediction of the neural process theory is that it's going to be the rate of change in in beliefs basically that is what's going to be measured as as an erp um you have anything further you want to say about that chris no no not at all i mean one thing i just want to highlight is like the the only difference between phase one

and phase two is a very slight difference in the precision of the a matrix that we um took to basically be the addition of some uh some slight focal attention um and but you can see like just looking at the figures like it really made very a very minor difference but it still bumped it up enough above the threshold just enough for it to basically go from 50 report to basically 100 report just also one point on the erp or the event related potential it's defined on the bottom left of the slide it's like a population level measure of the electromagnetic activity in populations of cells in the brain and so it's kind of like on the left if something's passing under the radar the population is not changing in broad strokes but then when there's this event that's task relevant it transiently synchronizes the population to be doing certain things in a certain timing and then that's reflected by changes in the population level electromagnetism patterns reflected by what are called erps but there's a whole literature with fristen and many others on this topic yeah so basic erps are just depolarization of dendritic trees yeah which corresponds to the prediction error yeah which corresponds to prediction yeah cool so basically as prediction initially goes up and then goes back down then that would be the uh the predicted erp um so um and this um is kind of related to uh you know pretty standard neural mass models um that are used in the in the literature that make pretty similar assumptions right they assume that the average firing rate of a population can be treated as a sigmoid function so a function that looks like this of the average membrane potential um and you can do things like for instance shift this function to the left or the right based on things that uh control the uh the firing rate threshold um you know which could relate to inhibitory interneurons or you know there's a bunch of different neural mechanisms involved but basically this the point is just that um this idea about where you get um firing rates um from average membrane potentials is is not kind of out of nowhere it's something that's kind of used as a standard assumption in um in uh these sorts of models and one interesting note is the soft max that sigma that we use to kind of renormalize probability it's actually utilized in artificial neurons in neural networks as also a probability renormalizing trick so we're thinking about it in kind of like the intersection of the physical thing and the instrumentalist approach realism and instrumentalism and we're sailing our active inference ship together through there yeah so so that's and and you know later chris will show specific simulations based on an actual task that will produce predicted erps that are consistent with existing literature um when he goes through the code um in the second part of this uh this session um but so i'm gonna sort of stop talking about the state prediction errors at this point which again chris will pick up um in his part the second thing that's kind of commonly talked about as part of the neural process theory in inactive inference is this idea of

expected free energy precision μ being associated with dopamine so let's say right from the start that to be technically correct we call this expected for energy precision in the literature it's often just called policy precision but that can be slightly confusing just because the precision of the actual posterior distribution of our policies right this bolded π here corresponds not just to the precision of expected free energy g but also to f and to e μ so there are ways in which the expected free energy precision this gamma thing that scales g μ you know could be precise or imprecise while the actual posterior distribution of her policies could still be precise or imprecise in a way that differs from μ the precision of g yep so you can be not sure what to do but have very clear understanding of two policies and their implications it's two separate levels of uncertainty no it's not quite that it's my life yeah it's not yeah like like so so a good a good example would be for instance so μ in the μ you know in uh you know the paper uh you know that i wrote with casper μ and others on the on deeply felt alpha so deeply felt affect so the idea about μ changes in emotional states or affective states μ relates specifically to this uh beta term here that controls the expected for energy precision μ now when expected free energy precision goes down μ then that will often correspond to right a negative or when expected for energy goes up sorry then that will correspond to a decrease and expected for energy precision μ but uh or a difference between f and g μ but uh but there will be cases where gamma will go down meaning the the model becomes less confident in its beliefs about uh its ability to minimize expected free energy essentially it decreases its beliefs its confidence in its action model but there are cases when that can happen despite the fact that the actual posterior additional distribution over policies becomes really precise so so a kind of nice example of this you know say you're kind of walking around in the woods and at the moment you're really confident in the walking around policy right so there's a precise distribution over you know i should be walking around but now you unexpectedly see a bear μ now immediately you're going to switch to having a very precise poster posterior distribution over the runaway policy so you're going to have two policies two distributions of our policies that are both very precise but they also but they change rapidly so the the policy that the that the model was confident in before is very different than the policy that the model was confident in after so despite that the posterior distribution is still over policies is still precise that things very confident what to do μ gamma will still go down μ so the thing will feel negative like in motion despite being very confident and running away so the thing that the thing that's driving that is f i think that's important to put in or to say if you want to cash it out ryan oh well i mean yes i mean just in the sense so f is f is going to be so based on a

new observation f is the u_m is the kind of evidence for each policy at the time of the new observation so if f strongly contradicts g then that means that the distribution is going to change in a way that suggests that you ought to have less confidence in g and u_m and we can we can i'll show a specific example of that below u_m when we actually show simulations of the u_m you know postulated dopamine responses but so the idea here is is that dopamine phasic dopamine responses correspond to changes in this parameter β which controls γ which is the expected free energy precision u_m and the way that these updates in β this equation here just corresponds to the initial β so the non-bolded one here plus this thing so the posterior distribution of our policies minus the prior distribution of our policies so basically how much did beliefs about policies change and then dotted with the negative u_m that was that you know that correspond to that trial and then minus the previous uh β which allows this thing to just continually iterate u_m but because the change from posterior release or policies from prior beliefs or policies u_m just adds f basically the prior over policies we don't show it here is just $\ln e^{-\gamma g}$ so the f only comes in to turn the prior over policies into a posterior policies u_m and so this change is driven by $f u_m$ and so the more that f is consistent versus inconsistent with $g u_m$ is what uh is what controls the the agent's sort of changes in confidence about about its g estimates u_m and so changes in that are what are expected or proposed to correspond to phasic dopamine responses and then the kind of tonic level of β is also proposed to be associated with tonic dopamine levels so if i can just very briefly say u_m so essentially γ is a γ is a a rate or bright β sorry is a rate parameter from a γ distribution and essentially we get you get this derivation so $\beta \dot{\beta}$ that that technically that's a that's a temporal derivative right and so this isn't exact it's a little misleading kind of because this actually is a variational derivative of γ with respect to variational free energy if that makes sense so the idea is is that updates or changes in essentially this β value will minimize variational free energy and if you want to see like the so the actual u_m the actual derivations of that get kind of kind of gnarly but if you're interested if anyone's interested i would just say u_m go and read and the appendix of ann sales paper that's in plus computational biology on the locus cerealia cyrillus and u_m learning rate and prediction error u_m where they show motivations yeah ana's yeah anna's anna's the appendix and this paper is like actually really great for this because plus cb uh basically like required it looks like that she spell like every little detail about deriving everything u_m in active inference so it has a really great paper both in the main text and in the appendix for showing kind of exactly where you get each of these things u_m uh yeah like pretty much as clearly as i've seen

it anything else to add on the neural process front because there's some questions and then we'll head into the second section but this is awesome uh there's a lot more actually let's keep going then then so yeah we'll do more neural process theory however much and then we'll be taking questions we'll have a intermission with a few questions that are really nice and then christopher will launch into the hierarchical modeling so so to actually give examples here because you know everything we've said about the about dopamine and beta updates is um still pretty abstract um so you know in the in the tutorial we give specific numerical examples of this as well um um to show and give a kind of like geometric actually um intuition um about this um and um so here you can see so so here we show right that the prior over policies this π_0 like i said is just the soft max of $\ln e^{-\gamma g}$ and then the posterior is the softmax of $\ln e^{-f - \gamma g}$ which we showed and so and these are the equations for quote unquote dopamine phasic dopamine responses um and so if you just take a specific really simple example so let's say there's no hat so let's say that $\beta_0 = 0$ and therefore $\gamma = 0$ and the initial beta posterior those are all just one so we're just starting out you know totally flat basically um and and we'll say that the agent has no habits um so there are two possible policies here i should say so the agent has no habit so it's just one and one for counts over uh policies now let's say that g is 10 and nine so in other words there's slightly higher expected free energy for the first policy than the second policy so now let's say that f the agent gets a and the free energy for each policy corresponds to 20 and one so as you can see this is still consistent with um the 10 and 9 right it's not like contradicting the um the you know policy one still has higher uh free energy than policy two um so when you calculate uh π_0 using this equation and these numbers up here you end up initially having a probability over policies of 0.218 and 0.728 so policy 2 is the one that is has is uh more likely uh given the expected free energy because it's lower right expected for energy is nine um now the posterior policies after we put in f just ends up being zero one um and so if you take the difference here so we're just called π_{diff} so $\pi - \pi_0$ that ends up being negative 0.218.218 and then if you do this a dot π_{diff} with negative g um and these two betas just cancel out because it's just one minus one and so you end up with 0.22 and then if you compute the update then uh the yeah what the change will be then γ ends up equaling 1.28 which went up from one um which means that you'd have a positive uh phasic dopamine predicted positive phasic dopamine response um and a way to kind of think about it geometrically is um whether or not the vectors corresponding to uh π_{diff} so the difference between π and π_0 point in the same direction as negative g um because that implies that they're uh consistent with one another that implies

that essentially the g was on track u_m and that's worth a re-run so what is this geometric space and then what would it mean if the vectors were totally correlated and they were like pointing the same exact direction versus here they're almost at 90 degrees or what would it mean for different arrows to be pointing in different directions so these are the vectors corresponding the vectors here in g and the vector u_m that corresponds to π minus π zero one so this is so this is and i should say that these are scaled u_m so that they're similar lengths but the same directions because otherwise one arrow would be way way longer than the other and the kind of point isn't clear wouldn't be clear but so basically π diff right is negative 0.218.218 so it's pointing left and up so negative x positive y u_m whereas g is u_m negative g would be negative 10 negative 9 so it will be down and left u_m and so the the general kind of u_m intuition that's supposed to be given is that u_m when these two vectors point in the same direction where that just means that the angle is less than 90 degrees u_m then uh the γ update will be positive u_m if for instance instead we do a different example now the only thing we're going to do here is we're going to change f we're going to change off from 20 and one to or no sorry we're gonna change g so g is going from 10 to nine to only being one and nine so in other words while in this case the first policy at higher expected free energy now the first policy has lower expected free energy now if you run through the exact same set of steps to solve the equations you're going to end up with a γ of 0.14 which means there's going to be a negative there should be a drop in dopamine in other words the agent should become less confident in its uh g estimates and you can see that if you look at where the vectors are pointing in this case u_m then they're then the angle difference is 128.66 degrees so more than 90 degrees so therefore they point in different directions therefore they provide evidence against g u_m if uh if that uh makes sense yep let me try with the dopamine so we're thinking of dopamine as a tracker of confidence it has many roles and colombo and others have talked about pluralism in dopamine so we're not this is not a lecture on dopamine we're thinking about dopamine as like confidence and we're headed in a direction we're on our road trip and the dopamine means coursing through our brain and we want to know how confident we are that we're on the right path as new observations are coming in now as we've seen from the earlier sections we're doing a lot of matrix math and just like a vector is a simpler version of a matrix and a tensor is a more complicated version of a matrix they're all kind of in the same math space and a given vector from computer science it's like two numbers that are in a column but also a vector has this arrow interpretation so if someone says the coordinate 3 comma 3 it's like a vector from 0 0 pointing to 3 3. so lists of numbers like a computer science vector is

basically the same thing as a vector with a star and a head at the at the end so it's physics and computer science and we're thinking about vectors and matrices and tensors and then we're laying out after scaling and renormalizing which direction these different vectors are pointing and then it's kind of like it are we headed on the basically the same path is the the vector of the evidence sort of pointing us in a similar direction that we're already believing in or is the new evidence coming in like a tailwind or a headwind that's kind of surprising us and it's not in line it's not headed in the same direction as what we're already believing and then this meta with a gamma the hyper parameter is about how much reliance we're putting in to g versus our observations yeah so i mean another kind of a nice a nice thing here is that so as as gamma goes down right so if you look if you look at this posterior you know distribution of our policies equation as this gamma value gets lower right so as the agent becomes less confident in its in its action model more or less it's less confident that its model will be good at minimizing expected free energy right as that goes down then g contributes less which means that f and e contribute more if an agent has built up really strong um then its beliefs in its model will action if gamma is high but if gamma's really low then the agent will just kind of choose whatever it's uh habitually chosen in the past so habits will kind of take over if it's not confident in its model um which is kind of a nice way to think about it has some kind of analogs to like model based versus model free um uh algorithms and reinforcement learning for example that um again we could think about in terms of actually making decisions based on you know explicitly simulating what's going to happen in the future if i make one choice versus another versus just doing what's typically worked in the past um is another kind of way to think about what gamma is arbitrating um did you uh do you want to say anything else chris no i was just going to say there's some really nice numerical results um so i think i've forgotten i'm sorry if i've gotten the name um i think it's thomas fitzgerald is the lead author on this but um they have some stuff just showing that related schemes i think that the equations have changed a little bit since then but they're pretty much the same um that this scheme gives very very similar answers as something like temporal difference learning which is a very simple model-free algorithm in reinforcement learning um and that's also i think the the gold standard model of dopamine phasic dopamine at the moment ryan correct me on that but no i mean yeah i mean it's like temporal the temporal difference account of of uh reward prediction error is definitely yeah treating phasic dopamine responses as reward prediction errors in temporal difference learning models is definitely a very standard and kind of like widely supported view um and this i mean the idea is that this this can um this sort of um this sort of active inference

model of dopamine um is kind of saying something different it's not saying it's a reward prediction but it's something that will have the same kind of dynamics as reward because if you think about it so say you know i'm walking along and i'm not expecting a reward and then all of a sudden i get a cue makes it so now that i expect now i expect um reward um so you know for instance or say like in like a simple kind of like experiment like a like auburn condition experiment conditioning experiment with like a rat or something you know it learns that every time it sees a light it's going to get a reward in five seconds um then the reward prediction area story would say oh you're going to get a real word prediction error when the rat sees the light because now it's going to get it knows it's going to get a reward that it wasn't expecting um so in this case though you'll get something kind of similar but for a different reason where you can think about the rat initially just kind of wandering around not being super confident you know in in any particular policy but then it gets that cue that it's going to get rewarded you know if it goes up and pushes the the lever right as a standard way this works um so so instead of it being a reward what would be happening is the queue is actually um unexpectedly making the rat a lot more confident in what to do right a lot more confident in the policy of going and pushing the lever um so you're going to get something that looks just like a reward prediction error at least in that sort of setup um but actually has to do with an update and confidence about what to do so so that's kind of in part that's one kind of intuitive way to think about um you know where similar looking dynamics come from but but i should also kind of be clear that i mean this is definitely not meant necessarily to be kind of a universal uh unifying account of dopamine you know at least at least as far as um you know i can tell a present there's lots of other things that dopamine does that you know look like they couldn't be explained by just this sort of model but this sort of model um looks like it's decent as a contender for making sense of things like um why dopamine doesn't just respond for prediction errors but also responds for events that are salient um for example um this kind of saliency versus word prediction or sort of thing there's a couple others but but um this is not necessarily meant to be a you know this is all dopamine does um but anyway um so uh so just making that clear um so now the kind of last thing to show and this is the actual figure in the paper that just shows both of these things is that um if you do the actual variational updating where just beta continually changes then the uh the actual kind of gradient like that the changes look like this where the thing would start out for instance at um around at a certain value um and then for for this case right so it starts at um you know down coming from one and then it will gradually go up and converge um after uh 16 iterations is the what's set in the

code um to that posterior value over what gamma ought to be whereas in this case it will drop coming up from one down and then converge over those iterations onto um this value of 0.14 so it's not as though in the neural model this is kind of happening with one little step it's it's kind of a it's a you know it's a convergence um that you can think of as related to a kind of prediction error um essentially kind of like an expected free energy prediction error uh some people have talked about it that way um in previous papers and for those following along in the code would this be in the efe precisionupdating.com yeah yeah yeah thanks so yeah there's a yeah to play around with this exact this this exact example um as well as ones where you can specify uh bigger policy spaces so like like when the agent has like five policies to choose from instead of one um you know which is a little more realistic it's just um it's hard to show uh vectors um sorry to illustrate those in like five dimensions um so we went with two for this simple examples um but yes um in that efe precision updating you can reproduce these results and then try changing like private beta values or changing g and f um to see to get more of an intuitive sense of the dynamics um and so like in the um in the eq in the the um what you might call it when we were showing in the explore exploit test model last time um these sorts of simulated simulated behaviors right where the agent either chooses to take the hint and then go left or right to get a reward oh it just makes or it's just risk's risk seeking and just goes left versus right automatically instead of taking the hint first um in in the previous session we went over this and but we kind of ignored this bottom right section on expected uh free energy precision and these dopamine plots but now that we've kind of walked through that a little bit um you know we can kind of come back to that and see that in this case um the agent started out in the beginning and it chose to take the hint um and then it knew um whether to choose left uh choose the left machine or choose the right machine so you can see what happens here is when it takes the hint then it becomes a lot more confident about what to do and so it's at that point that you simulate a big jump in dopamine right so in this case it's not because it's got a reward immediately it's because it got a cue where it learned what to be able to do to get a reward um so then it chooses left and then it observes a win here at the third uh time point or this third column um so so this is kind of an example um and then in the learning simulations that we showed last time as well before we leave that slide could i just jump in real quick and i had a quick clarification about that figure um so in that middle panel for win lose and you look at that matrix it actually looks like if that's the expected probabilities for win loss at any given uh time epoch one two and three um he actually it actually looks like in this case the way that the simulation goes it's more confident about winning on this time step when it knows to take

the hint and then after it's taken a hint um it becomes less confident about winning is or is that it's actually to do with the fact that those aren't so they're probabilities but they're not probabilities in the way that the rest of the modulus probabilities so these are these are preferences where we model preferences as a probability i see i see so that has to do with more of the you know it's had more update it's had more observations at time three and so as you're going since you're it's an accumulator um in that optimization process is that no it's just it's just so when you're trying to encode essentially what's rewarding for the agent right you're just setting you're just saying that it has a particular distribution over each possible set of outcomes for each outcome modality and that distribution just encodes which observations are more preferred than others and formally that's just a fixed uh probability distribution over the different observations you might get where the higher the probability is quote unquote the more the agent prefers that observation um so this is so this is just saying like for the first output modality it's just all gray here because the agent doesn't prefer a hint or not a hint um and then the third one here observed action it doesn't prefer innately to observe itself do one action or another but in this middle one observing wins versus losses versus null just not observing an outcome yet um at the first time point so the first column it just does it doesn't have any preferences at the second time point so this middle column the thing has a strong preference for win so this black one which corresponds to high probability it has a strong preference against losing observing a loss which is a very low probability and then the null state is just this kind of like intermediate thing it's not it's not bad like a loss but it's not good like a win and so the person that shows the null there or there is is because it shows a hint because of the epistemic value of the hint i should say yeah so it so basically what's happening is if you remember that the uh this distribution is a little lighter at time two because the value of winning is higher at the second time step than the third remember because it can win four dollars if it tries to get us it just chooses one right away but at the third time point this is a little less precise it's a little flatter which is like looks darker gray because the black only corresponds to winning two dollars instead of four um so the difference between winning and losing isn't quite as stark i think okay yeah um so but the point is it's not actually getting a big bump in dopamine when it wins you know it's getting a big bump in dopamine when it observes the hint because that's when it becomes confident in what to do to get the win um and that's to re-emphasize that we're doing this trajectory of policy which is what ties it to the path integral as so at each time point that it's being evaluated at it's in the past and in the future and it's conditioned on policy so it's almost like the reward if you always expect the envelope has

the paycheck if you get it and then you open it it's like neutral with respect to what you already believed about the way the world worked and you'd get a negative if it was less than expected and a positive if more than expected and so in that way the reward of what is initially just pure stimuli that could be rewarding or thought of in that way gets moved up that's kind of like reinforcement learning or reward and that's why we're in this area of thinking about stimuli and policy and rewards and risks yeah i mean the idea is just that in in that context uh these dopamine responses the simulated dopamine responses here corresponding to changes in confidence about what to do or changes in confidence that your model will give you the right answer about what to do and uh and that will look that will have the same dynamics as a reward prediction error from a queue that predicts reward so so the dynamics will look like a reward prediction area even though in these models it corresponds to something different um so it's kind of like in behaviorism everybody can agree on the behavior we agree that if we train up the animal it learns to do this after x um experimental paradigm is carried out and then now we're talking about internal we're thinking either at a neural process level like which regions of the brain or computationally like what is happening and then here there's a common trunk with the predictive processing the reward prediction error all of that and the active inference models we're talking about which is and reinforcement learning which is that over time the reward gets moved up to be associated apparently more with the queue than with the actual stimulus delivery but we're taking a point of divergence in that we're adding in a few different pieces that differentiate an active inference free energy gradient descent driven policy selection just q association even though they do have some similarities this is like a generalization or a much more nuanced way to approach similar situations yeah and you know ultimately right the hope is to find situations where active inference and reinforcement learning models are going to be going to make different predictions right about what dopamine will do um and uh because that's where you can actually say you know which model has more you know empirical support um and um yeah thomas fitzgerald i think it was the paper that chris was talking about before you know he did show there are interesting findings where when you get rid of dopamine when there's dopamine depletion there's some kind of lack of dopamine in the system for one reason or another um that doesn't actually get rid of uh reward learning like an agent can still learn about rewards despite um the dopaminergic function being impaired um you know which makes it look as though um you know dopamine uh might not be serving this reward prediction error function at least if you think that reward prediction is necessary for reward learning um and um and so there was he showed some simulations

again with a slightly different formalism than the current one showing how you can get the same um as you would expect in a reinforcement learning model but it explains why you don't need dopamine to do the reward learning um but uh but like i mentioned the formalism isn't exactly the same as it now because that was a slightly older paper um so you know i think there's just a lot more kind of empirical work to do to try to kind of find uh you know differential support for one model of dopamine or another very uh any other any other uh comments chris um no i think just looking at the time um do you want to kind of speed through the slides and then you can get onto the stuff yeah okay so yeah i mean i can skip i can skip or to skim over a lot of this i mean the last thing i was going to talk about is that this isn't used as much um because it's not actually implemented in this way in the code but carl has made this point that you can also implement minimizing expected free energy i'm using a different kind of prediction error called an outcome prediction error and that just corresponds to the different elements in expected free energy so which is here um here is just minimizing the expected difference between the preferred outcomes c and the outcomes that you expect under a right so um under some policy uh for under some policy the state for that π um multiplied with a will give you the expected outcomes of a policy so you're just trying to um this prediction error minimizing this prediction or just corresponds to finding the policy that min or that minimizes the the difference between basically what you want and what you expect given what you choose to do and the second term is the ambiguity or information seeking term and it effectively you can think about it as the expected difference between beliefs about states before and after a so you're trying to find basically um a difference that like uh the state that maximizes how much information you gain about what state you're in um but formally it's the it's the entropy of the likelihood distribution for a state or how kind of flat or uninformative the distribution of expected outcomes given that you're in one state versus another um and in the paper we um we uh go through uh numeric numerical examples of um computing these sorts of outcome prediction errors as well um but the the basic point again i won't go through them in detail is just if you have some policy that is going to generate the state uh state one with probability point nine and state two probability point one and your preference is perfectly precise over observing outcome one um and that's your likelihood mapping then um policy whereas policy two is gonna is expected to um just uh generate 0.5.5 over states um then um and the outcomes that are associated with those states look like this then what you're going to get is a policy one is going to end up if you do the math um out policy one is going to have a value of 2.4 for that for that term whereas policy 2 is going to

have a value of 7.3 so this difference is going to be a lot bigger and therefore there's going to be more outcome prediction error and so the the policy that the first policy that's going to generate the outcomes that are closer to what you want is going to minimize that prediction error more and so again we just show examples we also have some examples of of this part but the basic idea here is if this is your likelihood function then state one here is kind of imprecise in what it predicts right it predicts one outcome with 0.4 and another outcome with 0.6 whereas state 2 has a more precise it generates outcome 1 with 0.2 and outcome 2 with 0.8 so the agent would learn more if it moves to state 2 because the outcomes are more informative and so basically these work examples just show how um the agent should to minimize outcome prediction or the age shouldn't the agent should be driven to uh choose to move to state two um and so um finally and i'm not going to go into this at all but we also show this um panel here that corresponds to another kind of element um that's been proposed in the neural process theory that has to do with model reduction and sleep and the basic idea is that sleep allows you to search for other models that you might you might entertain that can produce the same experiences you've had during the day but can do so in a simpler way so it's finding so sleep is basically still trying to minimize free energy but not with respect to new observations but just with respect to your model based on the experiences that you already had earlier and that corresponds to internal dynamics that can essentially prune away or reduce the weights of synaptic connections that are contributing less to explaining um what you did so essentially kind of noise uh like like if there's a bunch of kind of coincidental um uh you know connections between different things you observe then there's some kind of noise in what you've learned and so you can during sleep you can reduce that um and here's just a few papers that have talked about that and shown simulations um in case people are interested um but so now moving on to what chris is going to talk are hierarchical models and the idea here is you can just take the same exact structure of the model that we've shown before um but instead um but put a layer kind of below it that's the same kind of thing but instead of the observations being actual outcomes the observations are just whatever the states the beliefs over states are at the lower level or the observations the observations for the second level are just the posteriors over states at the first level and so that just looks like this where you see here that you still are selecting policies at the higher level although you can also select policies at the lower level but states at time two here just generate states at time one and states at time one or states of sorry states at level one act does the observations for level two as you can see necessarily these operate over a slower time scale as well so for instance at time one level two

the first state state of time one is gonna generate the uh first state at time one at the lower level but then the lower level is gonna move to its beliefs at state two over a faster time scale and so this whole thing is going to repeat at the lower level so the thing is going to infer posteriors over states based on two observations at fast time scales and then the posterior beliefs over states at the lower level are then going to propagate up and act as the observations um for state one at time one at the higher and then that's going to transition through a second level b matrix to beliefs over states at time two which then is just going to provide priors to a second trial to the start of a second trial at the lower level which again operates over a faster time scale so essentially this whole lower level model completes its belief update um before the higher level transitions to its next state um and in the neural process theory it really just involves taking the first level just kind of tacking on another level of neurons on top and making the observations of the second level the things that drive the and the states as just the posteriors over states at the first level so it's just kind of repeating the same thing but at a higher level in treating the uh the interactions between the second and the first level the same way that observations feed into the first level um and so if i raise a question on this slide from the chat someone asked can you please explain the relationship or difference between models based upon deep active inference versus sophisticated inference and how is that related to the current discussion around dopamine signaling whether it's phasic or non-basic so two questions one is what is deep active inference and how is that related to sophisticated active inference and how are those related to what we're discussing now about dopamine and i have i thought if but you can go ahead i can answer all these very quickly this is perfect answer go for it so deep active inference can kind of correspond to two things uh you can talk about it in terms of using deep neural networks to parameterize your your essentially your a's and b's a and b matrices or your policies uh that would be kind of like a machine learning application in when i say deep active inference what i generally and what other people have talked about like the paper that first came out called like deep whatever the impact of inference it was just one of these hierarchical models whereas deep in the sense that there's this uh temporally deep uh or temporally thicker one could say uh time time scale stacked on top and so then in terms of sophisticated active inference that's just to do with policy selection um and where you're essentially doing something that looks more like a tree search um so i would just say i don't know it's equivalent to backward induction and so you essentially instead of saying kind of propagating forwards on a tree you start at your terminal node and then you propagate backwards so you say given where i want to go how do i best get there and then propagate backwards uh and then i don't

really know what the phasic non-phasic thing means uh yeah yeah i'm not i'm not aware of anybody sort of saying anything about um if there's any change in the dopamine story in sophisticated active inference um in sophisticated act of inference you're still you're still obviously updating beliefs about policies when you make a um but uh but yeah i mean the main difference like chris said is just it's how you come to post your beliefs over policies by doing this sort of more thorough research where you have this additional element where you say you know given that i observe a thing at time two at a given branch and a tree what will my beliefs be at time two um and like like chris said it's equivalent to backward induction um and uh it's it's um anyway yeah so i without without again i don't think anyone's talked about this much but presumably you still make new observations that update your beliefs over policies as you move through time um and so i don't necessarily see how the dopamine story would need to change um but uh but yeah um and i will say in addition to what chris said um they're really um two kind of related uh meeting meanings um independent of the kind of machine learning deep neural network thing that chris talked about one is and like you said the standard one is talking about deep temporal models yes and that's just exactly what i am was showing here where the second level necessarily operates at a slower time scale than the first level because the first level has to kind of complete all of its time steps um before it can provide an observation to the higher level and then the higher level so basically each higher level state corresponds to a whole lower level trial you know so if you wanted to you could make a lower level trial have ten time steps but the posterior beliefs at that lower level at the end of this whole time step all those time steps would be you know what the observation is that you know provides evidence for the first state at the first the state at the first time point at the second level so and then second level states transition and then the second level state so that basically one trial at the higher level with a two time points each of those time points corresponds to providing priors for a whole trial at the lower level so you're going to have as many trials at the lower level as you have time steps in one trial at the so so that's what deep temporal models means it's just that the second level operates at a slower time scale than the first and provides priors for initial states at the first level um another another meaning that some people use like in the deeply failed affect paper is just d parametric models which basically just means the first out things that happen at the first level can be used as evidence for hyper parameters for the higher level um and like in the deeply filled effect paper basically we had second level states that corresponded to valence um but those were updated not by first level beliefs but by changes in beta at the lower level so when gamma updates were positive

right when the confident and expected free energy went up there was an ascending signal that then updated beliefs at the second level that corresponded to feeling better right so positive valence um and those also that state could also act as a prior on um what the beta update should be right how confident the agent should be um at the lower level as trials go forward so it can also have to do with with um with uh and it's quote unquote deeper parameters um as opposed to just deeper deeper uh levels that control priors over states at the lower level um but that's not as commonly used um by far the most common is just deep temporal models do you have any more to add on the neural process or can we do our switch over um no neural process is pretty done um the only thing is is that before chris starts because what we're simulating when we're doing this example in the code of a hierarchical model the example we're using is an empirical task called a local global or oddball paradigm so chris is just gonna i'm just putting up this slide so that chris can walk us through the um the uh design of this task so just before we cross over just while we take a breath i want to make one ultra rapid summary for um just contrast and then ask one question from the chat so the brief summary is we're dealing with the active inference framework we are within free energy principle we're talking about active inference that's why it's active now we've heard about a couple of adjectives deep sophisticated affective and even inactive inference and those have been in the titles of papers or they've been the titles and we heard about just very um concisely about what deep can mean whether deep neural networks or deep time and then there's a third definition of deep meaning like fully or radically parametric and that is interpreted within within the affect of valence framework within the paper that ryan just mentioned so that's deep and affective which are related then sophisticated is related to this tree surge counterfactual policy and state estimation we talked about the sophisticated inference in a previous stream and an active is highlighting more from an ecological or even a philosophical perspective about the embodied and enacted aspects of cognition so that's just a summary about a few of the flavors under the umbrella of active inference they don't necessitate a renaming of the lab because they're like variants or paper titles but this is the total umbrella that we're working under and then the general question from the chat which i hope that set it up for and then we'll address this question each person give a very you know short response that we launched into chris's when preparing a model for exploring free energy principle active inference in matlab do you often sketch out with sample bayesian calculations for each term to help clarify your thinking and track what will happen or in other words how do you go from thinking or specifying the generative model of even an experiment into what you do when you're talking like you go from this behavioral

example to the code how do you do that um i mean so for me honestly like when i'm actually setting up a model for a task um i don't really think about the equations hardly at all because you don't need to right i mean what you need to what you need to focus on is just what the necessary elements are in terms of states and observations um and policies um that are just in a task right so you have to think okay in a task what are the observations that the participant has right is there one possible observation modality they either observe this or that or are there multiple observation modalities like maybe they observe a cue and then they observe a reward right and so then you need two observation modalities um right you think about how many in each modality right is it just q or no queue you know is it just reward or no reward or is it no reward reward or loss you know so you just think okay what are the different types of observations and how many observations within each type just you know what does the actual participant observe um and then you have to think about the states right what are the what is the participants beliefs what is the minimal set of beliefs they need to be able to make a decision right how to get reward if it's a reward task you know so that might be beliefs about what uh trial type it is right or might be beliefs about what are the different choices i might make um right like moving into the state of choosing say option one versus option two um you know and then you have to think okay what are the different action sequences right like does the agent just push a button and get lucky you know reward or no reward or you know does the agent have to you know make a couple choices in a row you know just what are the actual options right and so i mean that's that's really what it boils down to is just figuring out what things do the you know what beliefs do they need to make choices what choices are available um you know i mean it kind of it kind of builds itself beyond that um i mean you have to you have to realize that the the update equations in active inference you know we've actually been showing are are very general right they don't change at all depending on what task you're implementing so you're not changing the equations all you're doing is using the exact same update equations on a generative model that can successfully simulate behavior on a task so i i mainly think you just have to think about matrix and vector so i can actually answer this question just by building or showing how you get from this task structure to the code in question perfect um and so just to basically build on that um all that you think that's is the structure of the generated model because the update equations are totally generic and unless you're doing something like fairly advanced where you actually do need to specify some type of addition to the generative model use a new type of distribution whatever then you would need to rederive the update equations and you're going to have some time on your hands using

taking all the functional derivatives and setting it equal to zero and figuring out what the answer is but i have never had to do that personally and i think very very few people in the active infants literature have had to do that okay so in terms of this task this is a local global or basic like auditory modified auditory oddball paradigm um it was kind of i think this was the first paper that used it although i'm not completely sure about that but basically they just have play you a series of tones and there are four possible conditions and what they're trying to manipulate is your level of or the the time scale over which your expectations operate because they're trying to get at expectations at different levels of the cortical hierarchy and so the idea is is that you can have regularities at two time scales so the first time scale is the local time scale and that is just the series of five tones say so it might and these tones can be two different frequencies so it might be beep beep beep beep beep beep beep beep beep that type of thing um and the idea is is that let's say you violate an expectation on this local time scale this might be something like the bottom row here um this would be beep beep beep beep beep beep beep beep beep and if that happens 80 percent of the time or 80 of the trials you would have like a global expectation or expectation over a long time scale that there would be a violation on the four on the fifth um beep if that makes sense but on a local time scale for neuronal processes say in primary auditory cortex or wherever that are operating over this very rapid time scale they are blind to this more global regularity and so you will get essentially what this will give you is expectations and violations of global expectations particularly give you a p300 so essentially that is a large positive potential over kind of like fronto-central electrode sites and that corresponds to a violation of regularities whereas a mismatch negativity correspond which is kind of a negative i think i actually haven't i'm not an auditory neuroscientist so i'm sorry about this if i get it wrong but broadly speaking it's kind of over association cortices in like all the tree association cortices not a hundred percent about that though sorry i should look that up um anyway and that will correspond that's a negative going potential that's corresponds to violations of local and so kind of i just the reason we use this is because it's kind of the simplest possible model that gives you some type of realistic dynamics and also kind of highlights the face validity of this computational framework if that makes sense um so any any questions would you like to add anything ryan before i pull out the code i mean i guess you know just just because i'm the one in control of the slide and the pointer here i mean i just just want to make sure that people understand exactly what what's corresponding to what you say so so basically the idea is just that at each short time scale right you could have each green one corresponds to a beep and the red one

corresponds to a boop right so at the fast time scale you could be surprised that there's a boop at the end of four beeps um for instance if most of the time for example it was just bpb beep then whenever there's a bbb boop then that's unexpected right so that's an expectation violation at a low fast time scale um but um what chris is saying is is that um you can also have a long time scale thing which is like how many times in a row is it bpb boop versus bbb beep right so one longer time skill pattern four bpb beeps in a row um and so if you come to expect four bpb beeps in a row then if you hear a bbb boop then your belief about the expected sequence you know of these four trials um is what would be violated or or not so at the fast time scale you can have violations for each series of bpb beeps at the higher time scale you can have i expect expectation violation about the sequence of bpb beeps um so that's so that's the idea is you can have long time skill violations and short time scale violations where the short test time scale ones correspond to the low level and the highest high uh the high level the second level of the model corresponds to the beliefs about the sequence of ppp beeps um so anyway so that's what's called global regularity versus local regularity yeah just for simplicity i just simulated the local deviation global standard condition so that would be um if you kind of look the third from the bottom on the right hand side of the screen imagine that um and the local standard global deviation um so that might be yeah exactly which is the very top um on the right hand side of the screen and so i'll just pull out the code um sorry okay so now we gotta switch screens yep cool nice very interesting so far thanks a lot everyone and great questions too so can everyone see my screen yeah it looks good it's coming let's see cool okay let me just okay cool um so i now cannot see the chat or any of the anything so if you need to get my attention just like you'll have to shout out cool okay or just ask us periodically yeah yeah yeah so at the first level um so there's two levels this model and at the first level it's very simple the model there are there is one hidden state factor and it has two mutually exclusive states um high and it just corresponds to the tone of the stimulus and just high or low then in terms of we then separate the generative model from the generative process here like we talked about in the previous code walkthrough and then as we're going through kind of in terms of the likelihood mapping it's again really simple uh it's just an a matrix so the a matrix is just an identity mapping where sorry these just to go through it very briefly to revise a bit the columns correspond to hidden states and the rows correspond to the observations so this is saying when you are in uh hidden state high tone you what is the probability that you will see a high tone well it's an identity matrix so it's probability one um when you're in state low tone what is the probability that you'll see a low tone and it's again hidden state one we can then separate the

generated model from the generative process and the reason we do that is because this idea of there being uh so the generative process it is going to be the probability of a high tone hidden state or high tone state out there in the world giving rise to a high toned observation is is probability but our generative model is noisy and if you want to actually simulate have an insular co-brain that's at all realistic you need to have some level of noise in that so what i did what we do there is we just run it through a soft max function run the a matrix through a softmax with a temperature parameter that will very slightly turn down the precision so if i just i'll just run all this and then if we just open up little a we see that's now not a perfect identity mapping but it's something pretty close by the way so the reason i do that is because one it's realistic but also if you want to get erps at all where erps are essentially changes in belief if you have a perfect identity mapping you'll never get erps because the agent is 100 confident all the time so you never have to change your beliefs if you've got perfect beliefs um okay then in terms of transitions um this is just the bees it is again a identity mapping so that's just saying these that's perfectly stable uh high tones don't turn into low tones low tones don't turn into hype tones and at this first level of the model this is now we essentially plug all those variables into our mdp structures that we saw last time and at the first level of the model the thing that we need to specify is this t how many time steps are going to be at the first level per time steps at the second level and we only want there to be one so i'm just going to set that to one here and again erp we need to set that to one this is essentially a resetting parameter mdp.erp is the extent to which you reset your beliefs at each hidden after each time step at each new time step and the reason so say you are modeling a human being wandering through a maze where your time steps correspond to say a couple of seconds in between things it actually you actually should to be realistic or to be kind of have a realistic model you should have a little bit of resetting in your posterior expectations if however in this case there's uh where on the scale of seconds or milliseconds then it makes absolutely no sense to have resetting going on in your posterior expectations so we set erp equal to one and um that will yeah the only thing the only thing i'll say here is this is this is something that you got to kind of watch out for because for whatever reason the default in the uh standard routines in the vbac the spm underscore mdp underscore vb underscore x script default is the default is four the default value for erp is four if you don't specify it and so you'll get pretty funny probably inaccurate um simulations for erps if you don't set it back to one explicitly so there's a time scaling factor for how things are playing out at the neurophysiological versus the behavioral time scale well i mean it affects behavior too it's just saying it's just saying basically yeah like

chris said it's just how much do your beliefs go back to baseline um could compare to your posterior beliefs about that time about the you know like after the first observation um so like you could be just as surprised or more surprised at time step two even if your beliefs at time step one already favored what was going to happen in time step two yeah in a lot of cases you just don't want that you want priors to carry over so just in terms you could have something like complete resetting which would be completely ridiculous but you could do it if you wanted to or you could have no resetting at all which is what makes sense in this situation because we want there to be complete carryover you can do model fitting on empirical data and ask whether it's a better fitting model to have a empirically a 0.6 or a 1 or a 1.5 could you could fit this you could you treat this as a parameter that you fit as well yeah yeah if you were fitting to like match erp waveforms or something yeah yeah i would definitely fit it um okay so this is now this this next part is where there are two key points in this where we kind of there is a separation from how we would do a first level model or what a single level model so we need to clear so essentially usually you'll just clear all of your a's b's and d's because you're going to reuse those matrices at the second level and then you'll save your mdp structure and run it through this little mdp check thing which should give you some nice little error like very usually pretty helpful error messages telling you if you've got some weird shaped matrices or i don't know if i uh if i just have a zero here and ran it for example it would tell me hey your b matrix doesn't make any sense um so then you'll then generally clear this mdp structure and save the mdp1 into this little mdp this other mdp structure here so then moving on to the slower time scale or second level beliefs so this is where things get a little bit more complicated and so there are three hidden state factors where these are all independent probability distributions the first is sequence type and so this this is essentially there are three sequence types where this corresponds to sequences of tones at the first level so there can be a whole sequence where it's just there's no boops and so that would be all high there could be all low so it's all boops there could be a high and then a low for example and there could be a low and then a high so this would be does that kind of all make sense and this correspond where this corresponds to different sequences of hidden states at the first level got it not all of them have to be used in each experiment but that's the whole state space yep exactly then this is this is actually where things get interesting so you have to specify when you build this type of models you have to specify a time in trial hidden state factor which is essentially the agent's beliefs about how things evolve over time and there's actually some really it seems kind of like when i first ryan first taught me how to build these things i kind of just just had the assumption

like oh yeah this is just like a this is something that you need to build for the model to work but it doesn't it's just kind of a bit of formal machinery that doesn't track what the brain's doing to my delight and surprise there actually is quite a bit of neurophysiological evidence that there are factorized representations of task phase in both medial temporal lobes and lateral prefrontal cortex which is really cool but essentially this would be something like the time in trial that so i'm at time point one regardless of what stimulus there is i'm at time point two regardless of what stimulus there is i'm at time point n or i'm in the final time point where i have to give a response um regardless of what my response will actually be does that kind of all make sense cool and then the last hidden state factor is just the agent's response so we make the to make this interesting so we could have just set this up as a hidden markov model and not had the agent give responses um which would have been fine just would have been a very passive model and just kind of to make this interesting and then so that people can read it and then build their own more sophisticated models and see how you would specify policies in a deep model i basically had the agent at the last time step so the sixth time step just say whether the last beep or boop in the trial was the same as or different from the previous the previous beeps or boops is that all clear so far yep yeah and then so the idea is is that these sequent types the agent has no prior expectations over all of them it's totally flat so when this gets when this d first d1 gets run to a softmax function this will just be a completely flat vector now but the agent always has a perfect prior belief that it is in state one and it also has the perfect prior belief that it doesn't know or it's just not it doesn't have any response prepared in terms of what report it will make again and what we do here is essentially we separate the generative model from the generative process so just put little d2 just so it's kind of not to confuse people with these different levels and we multiply what's this is actually really crucial so when you specify a d and separate it there's a little flag in the vbx script or the model inversion script that tells it okay you are now learning d and you'll get updates to d but sometimes you don't actually want to simulate learning you want in the very same way that you want to like simulate some process outside of learning or some process independent of other things in a lab when you actually run experiments sometimes you just want to isolate one process and simulation so to essentially turn off learning you just make the concentration parameters in these which are concentration parameters from dirichlet distribution like really massive and so the difference between say 100 and if adding one adding a count of one will essentially do nothing to the agent's behavior then if the concentration parameter is already at 100 which is why we specify this way yeah so if you think about it i mean depending on the

type of task if if the thing thinks that learning is going on because you're separating the then if you have low concentration parameter values then the uh then the epistemic value for parameter exploration will will make a big difference right the thing will actually be driven to move to states that will help it learn something about what those concentration parameters should be um so it can affect behavior in ways that kind of um uh cause problems for the kind of tasks that you're trying to model um so so one thing yes is that having them at really high values means that the agent just isn't going to do any seeking information about parameters because it just thinks it already knows them really well um but also it still is going to add counts to those distributions after each observation so like you know changing a distribution from like 100 to 100 2 100 to 101 the actual distribution is like still basically the same whereas if it was like one and one and you add a count so it's not one and two that's a very big change in the distribution um so both of those things together just kind of make it as though it doesn't think it it's learning anything cool okay um is that cleared everyone all good to move on yep so okay so then the next thing so i think the hardest part of doing active inference or building these models is factorizing the a matrix so these are all independent distributions right but they interact in the a matrix and so you have to that you have to think about that when you're specifying it and so what this one what this matrix says is so remember that our first hidden state is uh sequence type so this is just and our first outcome modality is just the stimulus and i'm just going to quickly dip down to see how these how this level is connected to the bottom level we have to specify something called a link function and this will just be a little matrix or a vector depending on many states hidden state modalities you have at the first and second level where the rows in this matrix correspond to lower level hidden state factors and the columns correspond to high level uh outcome modalities and so all that we're saying and so when you specify these things you actually need to be careful that say the number of even states you have at the higher level well sorry the number of outcomes that are possible in your outcome modality match the number of hidden states that you have in your first level at the first level does that make sense your second level of your model is treating those first level hidden states as observations so here the two observations are high tone or low tone and all this is saying so this this first column is high tone second column or high sequence second column is low third column is high then low and last one is high is low than high this is and we're specifying this for time step one to three so what we're doing here is we are going so we have a set up at a matrix where there's four hidden states corresponding to sequence type and two outcomes just to reiterate and then what we're going to do is we're going to loop the six

time points in the trial and loop over the order of states in the report and so then we're just going to step but um then basically i'm just going to set set set this up so that um the time that there's just an identity mapping essentially between hidden states and and i've done this i've looped this over all but this is actually not an accurate generated model because we want there to be an oddball at the fourth time step and so i just specify that here so i take this i set which corresponds to this kind of second hidden state factor and i set it equal to four and what you can see here is that there's now a change in the mapping so remember this first column is high second column is low third column is high low third and then fourth column is low high so we now see a change so this is then saying when you're in a high hidden state at the fourth time step you'll see a high tone when you're in a low tone at the same thing but it gets interesting or it gets a change you get a change from the matrix above at the fourth time step because when it's in a high low at the fourth time step it will it will hear a low low tone and at in low high you'll hear a high tone is everyone pretty clear about that yes and we're looping over all this one what j equals one to three in both of these is just saying we're looping over all possible responses so we're not making that identity that mapping between those hidden states the first sequence sequence um first hidden state sequence and in any way dependent on the response so it's independent of that so then the second outcome modality that we need to specify is just report feedback so with the agent we're going to have it make an action or specify a policy allowing it to make an action at the fifth time step at the end of the fifth time step so i'm sorry the sixth time step rather and we then need to kind of give the agent feedback there are three hidden states and this is no longer this feedback modality is not connected to the first level and so there are three outcomes null which is just we don't give the agent any feedback so it just has a totally flat preference distribution essentially over over null doesn't care for the first five time steps and then at the sixth time step we need to start messing with whether it thinks it's going to be whether agents thinks it's going to receive wrong versus right feedback so this would be saying something like okay so here you report that they are the same so this is response two so j equals two and i equals six correspond to the six time step so these where these correspond to what we're indexing into elements of these hidden state factors and then so the idea is then what this is saying is that at hidden state for one and two if you say they're the same remember this is all highs all lows if you then say it's the same you will get correct feedback and because this is high low and low high if you say it's the same will give you incorrect feedback then we see this is reversed when we say the agent is going to give the agent the option to report different so these two will be incorrect for high and low sequences

and correct for low high and high low everyone with me yep cool okay then again just like we did before we're going to reduce the um constant we're going to separate the hidden states in the um sorry the uh hidden uh process and reduce the precision by with a temperature parameter of two which is super precise it's just not a perfect identity mapping and we're then going to disable learning again because we're not interested in learning at the moment um by the way you do get fairly similar results if you do learning so i'm not kind of like uh cooking the books it's just that in terms of the tutorial it didn't really make sense to add an extra several lines on learning for a fairly simple simulation it's awesome and just to highlight that is a similar design move as when you mentioned several times separating the generative process from the model it allows you to have a cue that isn't always associated like high tone is the high tone they hear it perfectly but now it's a situation where they can't really tell if it's one shade of a color or another so it it allows a lot of expressivity and it will just be really awesome to in the coming months and years see and return to the code to understand exactly how different groups are implementing it to model different specific situations that kind of just go down these roads that you're just pointing out but we're staying on on the freeway but you're pointing out different exits that really are interesting yeah exactly so there's lots of i mean there's quite a few different like like in previous papers you know for instance like we've modeled things like noisy like noisy learning right so where for instance like in the generative process it is you're always being presented with one object right and that's like 100 true but the the a matrix and the you know you kind of soft max it with some precision parameter like chris was doing so it's it's um not very precise so it's kind of like for instance like seeing an object but having it be blurry you know or something like that um you know so it's how well do you learn uh from stuff even if what you're seeing is kind of imprecise or like for instance figuring out where whether a tone was present in some like white noise you know like into some experiments to do that kind of thing so yeah i mean these models are incredibly flexible um for whether you want to do just perception or just learning or just decision making or decision making with or without learning um so it's i mean that's part of why they're so why they're so uh nice yeah and so then just kind of moving on to the transition um so b b2 this is just so this corresponds to our first ident uh sequence of tones hidden state factor and this is an identity matrix four by four identity matrix because there are four hidden steps um this is just saying that our our sequence doesn't change to another sequence halfway through that makes sense right um now our things get interesting in terms of specifying the transition functions uh when we talk about the second our trial phase uh transitions and our policies so the idea here is

that we then specify this trial phase transition function so that it will kind of deterministically step all of the phases in the sequence so this is state one or row sorry columns r state hidden rows are will be kind of hidden state one hidden state two so what this is saying is that at time one a hidden state one transitions to hidden state two hidden state two transitions to hidden state three and so on and if i wanted to if i wanted to to simulate some infinite number like it's not actually infinite this is a finite horizon mdp but if i wanted this infinite sum or sorry simulate some very large number of these trials i could put the last one so this last trial or sorry when it's in the last report phase it will transition back to the first phase for example but i don't want to do that i just want to have the last time step in the trial as an absorbing state that it'll that the simulation will stop on and then uh simulate each trial as kind of a separate thing but just highlight that in case you wanted to do it if you wanted to like simulate um there are some ways of simulating kind of heartbeats and things like that where you specify a transition function that has an orbit built into it so then in terms of report this is this is the controllable hidden state factor and these it'll will add will now have like a multi-dimensional matrix um where the third dimension in the matrix just corresponds to basically the control state so this would just be uh i am in null and i am going to stay in null i'm not going to do anything the second one would be you can get from any hidden state you can chorus you can transition yourself into reporting same and this third one is from any hidden state you can transition yourself into reporting different then in terms of the policies there are six hidden six time steps so we specify that here we specify the number of hidden state factors so remember again there's uh sequence type uh time and trial and report modality so three then we've got two policies so these are the number of options that the agent will have and then these are deep policies so the agent which is consistent throughout the whole trial and so basically or this is to say the agent will be computing these over the course of the whole trial if that makes sense um and so all this is saying is that the agent stays in state hidden kind of action one which is just null it doesn't do anything it doesn't do anything it doesn't do anything can't do anything and at the last time step we allow it to choose and that just corresponds to kind of you can imagine not actually presenting the participant with in a psychophysics experiment or whatever not presenting it with an option to actually move change the key hit a keyboard or anything like that until the last time step of the trial um okay so last bit uh any questions so far no looks pretty good though thank you um then okay so then we've got the first c matrix this will just be a two by t or two by six matrix of zeros so if i just kind of hit go on this we can take a look at it [Music] whoops i'm going to hit run at some point soon we can check out

everything but this is just what the c matrix looks like so it this is the preference for the agent's preference for hearing a particular tone so this is just saying agent doesn't care at all it has no preference for tone then our report hidden state this is again it's going to be completely flat essentially except for at the last time step when it receives feedback it has a preference for not being incorrect and a uh preference for being correct um so i just uh just i just want to make sure people understand that that so the uh the brackets um you know we went over this last time but the brackets cor like are what say what outcome modality corresponds to what a matrix oh yeah so like c2 brackets two means that that corresponds to the in uh the a matrix uh with brackets too right so so that's how you that's how you know that this is the one that corresponds to observing feedback um whereas the first one is the one that corresponds to just observing tone so just just to make it clear that you use the bracket numbers to assign to the outcome modalities yeah and then so then that's basically we've built our mdp so we're just going to specify it so we've we've now got our earlier mdp structure if we remember so this is mdp one and this is the thing that we specified all the way back when we were talking about level one um and we then plugged that into our new mdp structure so we'll have mdp.mdp so this is the the markov decision process at the lower we then plug it into this our new mdp or this will just be a subordinate mdp and we have to specify a link function we talked about that before but just to reiterate rows here are or sorry i'll start with rows are lower level hidden state factors and columns are higher level observation modalities so all that we're saying here is that we want the low level observation modality or low level hidden state one to plug into observation modality one so the hidden states the first level will be treated but as observations at the second level um okay then we just plug in all of our variables that we saw before time our likelihood mapping the generative model for our likelihood mapping um etc etc and so c's these and our v which is our policies and then again we need to reset our decay reset parameter to number one then in terms of mdp that we can just specify names for our outcome modalities and our transition functions which will just kind of make interpreting the plots a little bit nicer and then lastly you just plug the whole thing this whole mdp structure into spm mdp check thing which will again give you some nice informative error messages which are extremely helpful um i still use that thing all the time um and then we can just run it and so let's do that right now and by the way so i i do have a lot of uh i'm just going to run it and kind of show you what it looks like and then i'm going to go through how i simulated each of the specific conditions because we did something similar to how we simulate ryan simulated learning okay cool so i'm gonna hide this stuff for now um slight spoiler um okay and so this is just i'll just

look at one of them sorry just so we get an idea of what's going on um so this is just hidden state sequence the sequence is the same the agent believes it was the same throughout the whole trial great that's true uh time in trial transition to the next point just as we expected it and then at the last time step because this was a um all kind of uh all high it just said they would um they were the and so then would it got the answer right and i think it's only sharing your uh it's only yeah it's only showing the matlab window we don't see that we don't see the plots oh really uh um okay let's see if i can goddamn it if you if you dock if you dock them you might you might be able to pull it up yep slide into the ide i can try um or share a whole spring one can you see the screen now yep yes now yes okay thanks now we get to the plot yeah sorry i was just like talking to myself then that sucks it was fun um yeah okay um so this again this is the stable hidden state hi this is our trend this is our kind of time time and trial thing stably transitioning from one point to another and this is our report hidden state which just says at the second time step the agent reported same and then it got feedback that it was correct cool so then i'm just going to quickly go through essentially how i simulated all the conditions because remember we want to simulate local deviation and global standard and local standard global deviation that makes sense so i'm going to simulate 10 trials where that's 10 collections of beeps or big boops in a row um and what we want to do is essentially have the agent where it is going to be doing some learning it's going to be learning a prior expectation over what sequence it should be in if that makes sense and so yes just to make sure they understand this so go back up to the diddy's so if you notice he's only multiplying d2 and d3 by a hundred yeah d2 brackets one the first the first up the first state factor is still just one one one so which allows it to learn one but not learn stay factor two and say factor three yeah exactly um and so what we should have at the end of our local deviation thing for example so we have two oh sorry wrong part of the code okay so for our local deviation global standard manipulation or simulation what we're going to do is just the first nine trials will all be a high low so it will be boop and then the global so then our tenth trial however is going to be a um is going to be the same because it's a global standard so the global standard the bbb boop but we should get a kind of local deviation and we should always get a mismatch negativity because there's the last one deviates from the kind of the local temporal regularity if that makes sense then in our local standard global deviation we're going to invert that so the first nine trials are all high low so again we've got bbb boops but at the last time it's time step there is a beep beep beep and when we do that what we we won't get a um we won't get any type of mismatch negativity or first level erp to that last kind of surprise response that last at the last time step

we will however get a global deviation because the second level is now really surprised that there wasn't a deviation at the first level does that all kind of make sense nice cool and basically what we do is we'll just specify n equals 10 this number of trials we'll put our we'll get our mdp structure that we kind of put together before we'll put it into our mdp deal so that will just be give us 10 mdps which are to start with all identical and then we're just going to choose to select the 10th one and change either have and mess with the hidden state factor so here we're not actually changing it's the same as the first 10 uh the first nine but in the second example we're changing it we're changing it from a local deviation to a local standard or um which will give rise to a global deviation and so now if i just i've then kind of i'm actually not going to for the sake of time i'm actually not going to go through all of the code for kind of just plugging this stuff in because this is literally just putting our structures into a plotting script and here things get more here is where i make some custom erp plots which i think are kind of cool but again i actually don't think walking through the code is particularly helpful i think it's just the type of thing like the way i learned this was i just spent like a day or half a day just reading through uh someone else's code and figuring it out um so the the only thing that i would make sure is that so when you um when you're doing this uh u1 underscore i'm sorry like it's just these are ways of indexing and pulling the uh the actual simulated um firing rates and erps um out of the spm out of the mdp structure after you run it through the vbx code yeah so these are this is just representing uh kind of the the most straightforward way of pulling those similar simulation results out of cells in the um in the mdp structure after it's run and we have a table in the tutorial i think it's table three that explains the structure of each of those cells and where each of those um each of those uh simulation results yeah exactly and so then i'm just going to hit run or kind of walk through what this stuff looks what these plots look like together um nice cool okay so these are my custom erp plots on the right and so i'm going to start start there actually because i think they're kind of nicer okay so cool so far left we have a or top top left sorry we have a global standard versus a global deviation and the way you calculate the mismatched negativity is you just subtract the global step the global deviation from the global standard and then you get this nice that's what we have on our top right we have this nice kind of negative going erp that looks exactly i mean i'm not going to show you this but if you want to satisfy yourself with it this looks like kind of for such a simple model this is it's kind of shocking how much this looks like a standard mismatched negativity so that's really cool then on our second one here this is our two this is our global standard which is in blue and our global deviation which is in red and then we can subtract

them from one another and what we can we can just look at kind of the um the we get a nice large positive going potential that looks a lot like a p300 so then in terms of our i'm just going to show you a couple of the local deviation local standard things um sorry i'm gonna close these and these just kind of clean up the screen it's nice though that someone can just download this and also an important note is installing all of the spm prerequisites maybe we could do a standalone in the future but once the packages are installed someone can just run the script and have also all of these figures be spit out and then it looks like there's enough commentary and hopefully these videos are enough of a walk through at least of early versions of the code where you can tweak different things and start seeing how it plugs into other things you might want to explore with the model but all these figures just come out in one batch and just fighting them away over there christopher yeah exactly so i guess one thing to say is one reason why i'm not particularly motivated to make a big standalone kind of thing is by the time even if i were to do that which i don't want to do but even if i were to do that by the time i'd written all rewritten all of the functions i would essentially have a package that is as cumbersome to download as spm but spm is written in a really nice way and it's super easy to install um and so it would actually be a lot easier it's just a lot easier to install spm that would be to go to my github say and put all of this stuff onto your laptop make sure all the paths work nice because they've automated a future supplement could just be start with a virtual machine and let's just install the prerequisites and just actually do a full clean installation could be a good point of reference but thanks for that point continue yeah i mean it's i mean it's really very it'd be very short and sweet to do i mean downloading spm like is you know it's freeware i mean it doesn't just takes a couple of minutes to download and really all you have to do is just once you download it is just go into matlab and set the path like that's it yeah i think there's a function called install spm so you go into matlab and run the run the function install sp um okay but returning to this so each of these points kind of everyone see my my pointer i'm hoping yes yes yep each of these pointers this is all a time step at the first level and so this is kind of local standard a global deviation right um so this will be we see there's a little erp each time there's new stimulus presented but there's not not a major one that makes sense because it's the same um and our this is trial 10 by the way i should say this is trial 10 so i i have trial one versus uh maybe i'll just maybe i'll just contrast all the trial tens because otherwise things get messy okay cool this is trial ten um and so this is trial 10 for local standard global deviation and so we see kind of first level erps really all no enhanced erps they're all the same but this last one because it this last or sixth erp right because it

disconfirmed a prediction essentially there is an enhancement in higher level beliefs when the model suddenly changes from thinking it was a high high low high high low to a high high if that makes sense and then if you compare that second level ERP to this one over here which is this local deviation global standard it's smaller than p300 there is smaller amplitude but at this first level because it kind of confirmed that there was a deviation at this first level you see this first level there's this massive like deviation response in um so does that all is that all fairly clear because i think that's just about all i have to show um i i do think this stuff is really cool it's kind of nice to chat about it so just to i want to be very clear that i'm understanding with the um event-related event-related potentials that you're showing that you superimposed and you got based off your hierarchical model um you could then superimpose those uh ERPs which are related to gradient descent on the variational free energy which is related to dopamine expression and as we showed uh way at the beginning of this video that relationship between membrane voltage and expected firing rate mean population firing rate and we're taking all of these things together to provide the superimposed traces of the local fuel potentials at any given instant um and we're showing how this oddball could play out uh conceivably by re you know with a simulated version of this model how it could be realized is yeah exactly the only thing i would say is that these these simulations have like nothing to do with the dopamine stuff yeah i was i was about to say the same thing inside okay with the exception of like dopamine having to do with how you choose so if we just like blew up the dopamine receptors or sorry this like in silico dopamine and just made it like we could make the agent behave totally randomly or we could make it behave in a super precise impulsive way lines sorry i shouldn't have said that uh impulsivity is a bit complicated um but we could make the um agent behave in a totally random way and choose random actions which would again have effects on the sensory sense on the sensations that the agent would receive so it would indirectly affect ERPs but there's nothing about like that gamma term that would affect uh state prediction error which is there so the primary could effectively turn off you could just turn off gamma right and and it wouldn't yeah it wouldn't make any difference to this exactly so so then just so that i'm getting it correct the more important thing then would be the membrane voltage which is related to free energy minimization and your schematic diagram of those excitatory imputative inhibitory connections that you're realizing in this model exactly yeah it's just it's just about rate of change and posteriors over okay totally so i'm just going to stop showing my screen at this point um so hopefully you guys can no longer see my screen nope we're back to just just you know cool well wow what a fun and session let's uh first just

really appreciate everyone for coming on because these are extremely didactic and it's really just something else to be walking through the code and seeing how many degrees of freedom there could be and also how real it can be when we actually specify it so any closing thoughts or and what can people look forward to for part four which is the final part before the semester begins for some of us yeah no i mean uh like i said i mean the last the last one is just the last section of the of the tutorial which is kind of where we're building to this whole time right is how to actually apply this stuff to empirical data and do real experiments uh can i ask a question on that is someone asks can we use this tutorial to construct a of metacognitive control of attention or meta awareness i.e modeling task-free conditions say while doing meditation for example so we talked a lot about task modeling how does this generalize yes i mean you kind of have to so i mean uh you know the first question is just how do you make that you know like really concrete right so i mean there's there's very easy ways you know like i've published a couple of papers and so has um you know several other people um doing something like cognitive actions selecting cognitive policies as opposed to behavioral policies and these are things like where you like selectively choose what to pay attention um you know or selectively choosing you know what to kind of hold in working memory or you know like things things along these lines that don't involve any kind of like over observable behavior but involve um selecting cognitive fully internal actions right like something like selecting ways of manipulating lower level representations things like that um so that's certainly possible but what you would need to do then is say okay you know what does uh meditate what does a meditative action look like right like it could be something like for instance making a like choosing a policy at a higher level that say like for instance like really really really reduces the precision of like an a matrix or something right if you wanted to completely you know like cease to attend to anything like in the external world you know for example um you know or for maybe the opposite right maybe like mindfulness you know has to do with actually paying really really precise attention to the present but like making your uh transition matrices super imprecise so essentially it's all about the present and like you're not thinking about anything in the past or the future you know or i mean i could i could think of lots of different ways you might do something like that but to really make it you have to turn it in to something task ish right you have to define actions you know even if they're fully internal actions you have to define the way they relate to perception the way they manipulate the processing perception um you know i guess like you have to ultimately boil it down to something really concrete with an exact quantitative structure but all the resources are certainly there to do it if you're creative

enough to come up with a regenerative model so i have a yeah i have two things to add to that so first is that i think there's a really there's a really cool kind of like early pre-print i'm not sure if it's been published anywhere it's just called towards a formal neurophenomenology of metacognition um lars sandoval smith is first author maxwell ramstead who i think has been on the podcast before right there he's the he's the senior um they actually do try and model metacognition and specifically in meditation and meditation context um i would just say one thing that i am personally skeptical about um i encourage everyone to do this kind of work but this is just this probably highlights my biases but i think if something is hard to study empirically that is to say it's very hard to study in terms of urine imaging for example how are you in creating like an event related design where you can get precise time locking to neuro between neuronal responses and cognitive processes for example if something's hard to do that in an experimental context it's going to be extremely difficult if not more difficult to do it meaningfully in terms of the generative model because i think these generative models are only meaningful and helpful to the extent to and other people are free to disagree with this um this is just very much you're going off on my own kind of tangent that's just to say it's almost like you need to get more rigorous it's no longer just like well this one thing happened with this one setup it actually demands a higher standard but things that it lets above that higher standard are very cross-referenceable very interoperable so very nice very nice statement yeah i mean i should i should say that like like i guess you can think about these models as you know beyond the kind of more specific applications we've been talking about is doing kind of two you know doing like a couple of like broader things one is is that you can use them for kind theory making theories precise right so for example like um so a lot of my past work like i focus a lot on like like theories of like emotion and emotional awareness and how that light still relates to like you know psychiatric you know like clinical contacts and things like that um but you know prior to doing a lot of this modeling stuff um it kind of comes out of these like fairly hand wavy conceptual review papers right where it's like boxes and arrows and you know like it's and sometimes in those cases it's hard to go from kind of the theoretical boxes and arrows um you know even if you assign say neural regions or systems to them it's hard to make those things really precise so that like it's very clear that one empirical result would actually confirm or just confirm it um whereas so you can take you know like a kind of conceptual you know somewhat hand-wavy model like that and then turn it into an exact generative model which can just make your theory really precise you know so so like one day you know it'd be really cool if instead of like review papers it's like review paper plus actually

here's a model you know like a precise quantitative model that we think comes out of unifying all the results as opposed to um less precise um theories one one other nice thing you can do with these models is just like have a total proof of principle someone someone might make a radical claim like there is no way that you can conceivably do x and then the response that is here is a model that does x um and that's useful right that might not be it might even not be an empirically tractable example like i think the dark room problem is an example of this right like you could say there is no possible way of designing an active inference that will escape the dark room and then you can build a model and say active inference agent escaping the dark room like that would be a really shitty boring experiment to run i hope one but it's a argument about argument by construction but especially when people have these edge cases or perceived to be edge cases but that's how philosophy is at the fringe and at the border and expanding and critiquing and improving all these different metaphors that have come up in discussing like 14 was what is the dialogue between science and the modeling and the actual quantitative components of it and the empirical work and philosophy so it's really just we're coming at that same question from a little bit of a different angle so i think this has been a great length and a great session ryan christopher max really appreciate all of you for coming on so that's going to be it for today but it will be our final session next friday at the same time at 9 00 a.m pacific on february 5th is going to be number four so thanks everyone for watching live and replay thanks again to authors and participants so peace out everyone yeah thank you bye thanks